MIPT & YSDA · AUTUMN 2026

Deep Generative Models

Lecture 4

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Recap of Previous Lecture

MLE for LVM

$$\sum_{i=1}^{n} \log p_{\boldsymbol{\theta}}(\mathbf{x}_i) = \sum_{i=1}^{n} \log \int p_{\boldsymbol{\theta}}(\mathbf{x}_i|\mathbf{z}_i) p(\mathbf{z}_i) d\mathbf{z}_i \to \max_{\boldsymbol{\theta}}.$$

A Naive Lower Bound

$$\log p_{\boldsymbol{\theta}}(\mathbf{x}) = \log \mathbb{E}_{p(\mathbf{z})} p_{\boldsymbol{\theta}}(\mathbf{x}|\mathbf{z}) \geq \mathbb{E}_{p(\mathbf{z})} \log p_{\boldsymbol{\theta}}(\mathbf{x}|\mathbf{z})$$

$$\approx \frac{1}{K} \sum_{k=1}^{K} \log p_{\boldsymbol{\theta}}(\mathbf{x}|\mathbf{z}_k), \qquad \text{where } \mathbf{z}_k \sim p(\mathbf{z}).$$

Variational Evidence Lower Bound (ELBO)

$$\mathcal{L}_{q,\boldsymbol{\theta}}(\mathbf{x}) = \mathbb{E}_q \log \frac{p_{\boldsymbol{\theta}}(\mathbf{x}, \mathbf{z})}{q(\mathbf{z})}$$

$$\log p_{\boldsymbol{\theta}}(\mathbf{x}) = \mathcal{L}_{q,\boldsymbol{\theta}}(\mathbf{x}) + D_{\mathrm{KL}}(q(\mathbf{z}) \| p_{\boldsymbol{\theta}}(\mathbf{z}|\mathbf{x})) \geq \mathcal{L}_{q,\boldsymbol{\theta}}(\mathbf{x}).$$

Recap of Previous Lecture

Variational Evidence Lower Bound (ELBO)

$$\mathcal{L}_{q,\boldsymbol{\theta}}(\mathbf{x}) = \mathbb{E}_q \log \frac{p_{\boldsymbol{\theta}}(\mathbf{x}, \mathbf{z})}{q(\mathbf{z})} \leq \log p_{\boldsymbol{\theta}}(\mathbf{x})$$

$$\mathcal{L}_{q,\boldsymbol{\theta}}(\mathbf{x}) = \int q(\mathbf{z}) \log \frac{p_{\boldsymbol{\theta}}(\mathbf{x}, \mathbf{z})}{q(\mathbf{z})} d\mathbf{z} = \mathbb{E}_q \log p_{\boldsymbol{\theta}}(\mathbf{x}|\mathbf{z}) - D_{\mathrm{KL}}(q(\mathbf{z}) \| p(\mathbf{z}))$$

Log-Likelihood Decomposition

$$\log p_{\boldsymbol{\theta}}(\mathbf{x}) = \mathbb{E}_q \log p_{\boldsymbol{\theta}}(\mathbf{x}|\mathbf{z}) - D_{\mathrm{KL}}(q(\mathbf{z}) \| p(\mathbf{z})) + D_{\mathrm{KL}}(q(\mathbf{z}) \| p_{\boldsymbol{\theta}}(\mathbf{z}|\mathbf{x})).$$

- Maximize the ELBO as a surrogate for log-likelihood:

$$\max_{\boldsymbol{\theta}} \log p_{\boldsymbol{\theta}}(\mathbf{x}) \quad \longrightarrow \quad \max_{q,\boldsymbol{\theta}} \mathcal{L}_{q,\boldsymbol{\theta}}(\mathbf{x})$$

- Maximizing the ELBO with respect to the variational distribution q is equivalent to minimizing the KL divergence:

$$\arg\max_{q} \mathcal{L}_{q,\boldsymbol{\theta}}(\mathbf{x}) \equiv \arg\min_{q} D_{\mathrm{KL}}(q(\mathbf{z}) \| p_{\boldsymbol{\theta}}(\mathbf{z}|\mathbf{x})).$$

Recap of Previous Lecture

$$\mathcal{L}_{q,\boldsymbol{\theta}}(\mathbf{x}) = \mathbb{E}_q \log p_{\boldsymbol{\theta}}(\mathbf{x}|\mathbf{z}) - D_{\mathrm{KL}}(q(\mathbf{z})\|p(\mathbf{z}))$$
$$= \mathbb{E}_q \left[\log p_{\boldsymbol{\theta}}(\mathbf{x}|\mathbf{z}) - \log \frac{q(\mathbf{z})}{p(\mathbf{z})}\right] \to \max_{q,\boldsymbol{\theta}}.$$

Variational Posterior

$$q^*(\mathbf{z}) = \arg\max_q \mathcal{L}_{q,\boldsymbol{\theta}^*}(\mathbf{x}) = \arg\min_q D_{\mathrm{KL}}(q(\mathbf{z})\|p_{\boldsymbol{\theta}^*}(\mathbf{z}|\mathbf{x})) = p_{\boldsymbol{\theta}^*}(\mathbf{z}|\mathbf{x}).$$

Amortized Variational Inference

We restrict the family of possible distributions $q(\mathbf{z})$ to a parametric class $q_{\boldsymbol{\phi}}(\mathbf{z}|\mathbf{x})$, conditioned on data $\mathbf{x}$ and parameterized by $\boldsymbol{\phi}$.

Gradient Update

$$\begin{bmatrix}\boldsymbol{\phi}_k \\ \boldsymbol{\theta}_k\end{bmatrix} = \begin{bmatrix}\boldsymbol{\phi}_{k-1} + \eta \cdot \nabla_{\boldsymbol{\phi}} \mathcal{L}_{\boldsymbol{\phi},\boldsymbol{\theta}}(\mathbf{x}) \\ \boldsymbol{\theta}_{k-1} + \eta \cdot \nabla_{\boldsymbol{\theta}} \mathcal{L}_{\boldsymbol{\phi},\boldsymbol{\theta}}(\mathbf{x})\end{bmatrix}\Bigg|_{(\boldsymbol{\phi}_{k-1},\boldsymbol{\theta}_{k-1})}$$

Recap of Previous Lecture

$$\mathcal{L}_{\boldsymbol{\phi},\boldsymbol{\theta}}(\mathbf{x}) = \mathbb{E}_q \log p_{\boldsymbol{\theta}}(\mathbf{x}|\mathbf{z}) - D_{\mathrm{KL}}(q_{\boldsymbol{\phi}}(\mathbf{z}|\mathbf{x})\|p(\mathbf{z})) \to \max_{\boldsymbol{\phi},\boldsymbol{\theta}}.$$

Gradient $\nabla_{\boldsymbol{\theta}}\mathcal{L}_{\boldsymbol{\phi},\boldsymbol{\theta}}(\mathbf{x})$

$$\begin{aligned}\nabla_{\boldsymbol{\theta}}\mathcal{L}_{\boldsymbol{\phi},\boldsymbol{\theta}}(\mathbf{x}) &= \int q_{\boldsymbol{\phi}}(\mathbf{z}|\mathbf{x})\nabla_{\boldsymbol{\theta}} \log p_{\boldsymbol{\theta}}(\mathbf{x}|\mathbf{z})d\mathbf{z} \\ &\approx \nabla_{\boldsymbol{\theta}} \log p_{\boldsymbol{\theta}}(\mathbf{x}|\mathbf{z}^*), \quad \mathbf{z}^* \sim q_{\boldsymbol{\phi}}(\mathbf{z}|\mathbf{x}).\end{aligned}$$

Gradient $\nabla_{\boldsymbol{\phi}}\mathcal{L}_{\boldsymbol{\phi},\boldsymbol{\theta}}(\mathbf{x})$

$$\begin{aligned}\nabla_{\boldsymbol{\phi}}\mathcal{L}_{\boldsymbol{\phi},\boldsymbol{\theta}}(\mathbf{x}) &= \int p(\boldsymbol{\epsilon})\nabla_{\boldsymbol{\phi}} \log p_{\boldsymbol{\theta}}(\mathbf{x}|\mathbf{g}_{\boldsymbol{\phi}}(\mathbf{x},\boldsymbol{\epsilon}))d\boldsymbol{\epsilon} - \nabla_{\boldsymbol{\phi}} D_{\mathrm{KL}}(q_{\boldsymbol{\phi}}(\mathbf{z}|\mathbf{x})\|p(\mathbf{z})) \\ &\approx \nabla_{\boldsymbol{\phi}} \log p_{\boldsymbol{\theta}}(\mathbf{x}|\mathbf{g}_{\boldsymbol{\phi}}(\mathbf{x},\boldsymbol{\epsilon}^*)) - \nabla_{\boldsymbol{\phi}} D_{\mathrm{KL}}(q_{\boldsymbol{\phi}}(\mathbf{z}|\mathbf{x})\|p(\mathbf{z}))\end{aligned}$$

Reparametrization: A Gaussian Example

$$p(\boldsymbol{\epsilon}) = \mathcal{N}(0, \mathbf{I}); \quad q_{\boldsymbol{\phi}}(\mathbf{z}|\mathbf{x}) = \mathcal{N}(\boldsymbol{\mu}_{\boldsymbol{\phi}}(\mathbf{x}), \boldsymbol{\sigma}^2_{\boldsymbol{\phi}}(\mathbf{x})).$$

$$\mathbf{z} = \mathbf{g}_{\boldsymbol{\phi}}(\mathbf{x},\boldsymbol{\epsilon}) = \boldsymbol{\sigma}_{\boldsymbol{\phi}}(\mathbf{x}) \odot \boldsymbol{\epsilon} + \boldsymbol{\mu}_{\boldsymbol{\phi}}(\mathbf{x}).$$

Recap of Previous Lecture

Training

1. Sample $\mathbf{x} \sim p_{\text{data}}(\mathbf{x})$, $\boldsymbol{\epsilon} \sim p(\boldsymbol{\epsilon})$.
2. Reparametrize $\mathbf{z} = \mathbf{g}_{\boldsymbol{\phi}}(\mathbf{x}, \boldsymbol{\epsilon})$.
3. Compute the ELBO:

$$\mathcal{L}_{\boldsymbol{\phi},\boldsymbol{\theta}}(\mathbf{x}) \approx \log p_{\boldsymbol{\theta}}(\mathbf{x}|\mathbf{z}) - D_{\text{KL}}(q_{\boldsymbol{\phi}}(\mathbf{z}|\mathbf{x})||p(\mathbf{z})).$$

4. Update $\boldsymbol{\phi}$, $\boldsymbol{\theta}$ via stochastic gradient ascent.

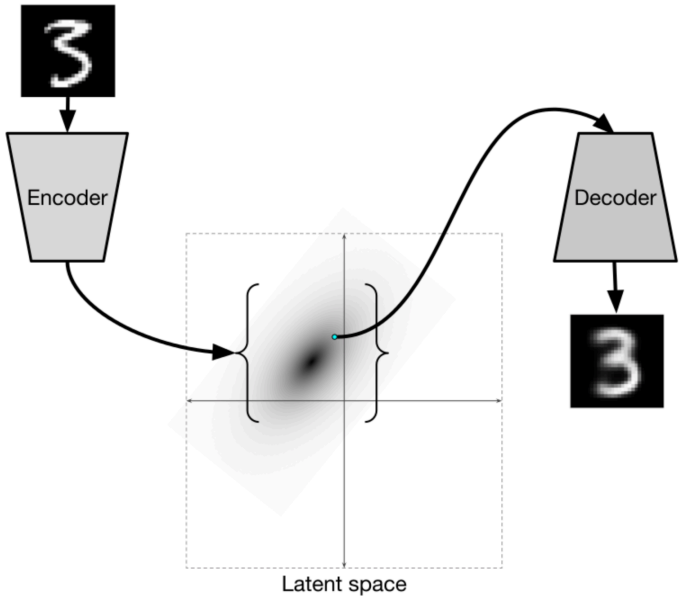
Sampling

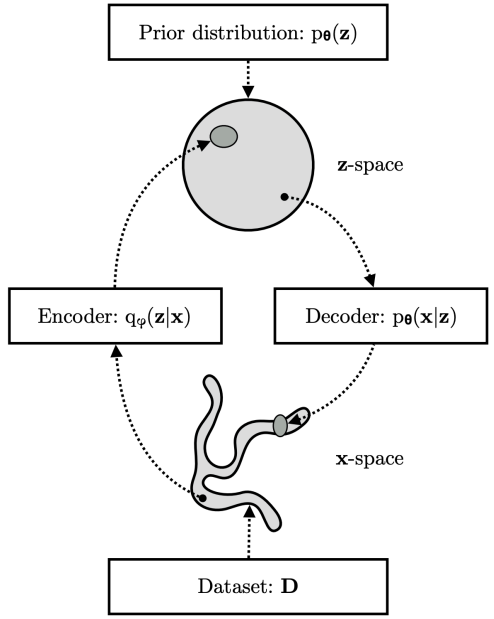
1. Sample $\mathbf{z} \sim p(\mathbf{z}) = \mathcal{N}(0, \mathbf{I})$.
2. Sample $\mathbf{x} \sim p_{\boldsymbol{\theta}}(\mathbf{x}|\mathbf{z})$.

Note: The encoder $q_{\boldsymbol{\phi}}(\mathbf{z}|\mathbf{x})$ isn't needed during generation.

Recap of Previous Lecture

$$\mathcal{L}_{\phi,\theta}(\mathbf{x}) = \mathbb{E}_q \log p_{\theta}(\mathbf{x}|\mathbf{z}) - D_{\mathrm{KL}}(q_{\phi}(\mathbf{z}|\mathbf{x})||p(\mathbf{z}))$$





Encoder $q_{\phi}(\mathbf{z}|\mathbf{x})$

Predicts $\boldsymbol{\mu}_{\phi}(\mathbf{x})$ and $\boldsymbol{\sigma}_{\phi}(\mathbf{x})$.

Decoder $p_{\theta}(\mathbf{x}|\mathbf{z})$

Predicts parameters of the data distribution.

Prior $p(\mathbf{z})$

Typically $\mathcal{N}(0, \mathbf{I})$; used for sampling.

image credit: http://ijdykeman.github.io/ml/2016/12/21/cvae.html
Kingma D. P., Welling M., An Introduction to Variational Autoencoders, 2019

Outline

01 ELBO Surgery and Optimal VAE Prior

02 Discrete Latent VAEs

Discrete VAE Latent Representations
Vector Quantized VAE (VQ-VAE)

03 Likelihood-Free Learning

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ELBO Surgery

$$\frac{1}{n}\sum_{i=1}^{n}\mathcal{L}_{\boldsymbol{\phi},\boldsymbol{\theta}}(\mathbf{x}_i) = \frac{1}{n}\sum_{i=1}^{n}\Big[\mathbb{E}_{q_{\boldsymbol{\phi}}(\mathbf{z}|\mathbf{x}_i)}\log p_{\boldsymbol{\theta}}(\mathbf{x}_i|\mathbf{z}) - D_{\mathrm{KL}}(q_{\boldsymbol{\phi}}(\mathbf{z}|\mathbf{x}_i)\|p(\mathbf{z}))\Big].$$

Theorem

$$\frac{1}{n}\sum_{i=1}^{n}D_{\mathrm{KL}}(q_{\boldsymbol{\phi}}(\mathbf{z}|\mathbf{x}_i)\|p(\mathbf{z})) = D_{\mathrm{KL}}(q_{\mathrm{agg},\boldsymbol{\phi}}(\mathbf{z})\|p(\mathbf{z})) + \mathbb{I}_q[\mathbf{x},\mathbf{z}];$$

Aggregated Variational Posterior

$$q_{\mathrm{agg},\boldsymbol{\phi}}(\mathbf{z}) = \frac{1}{n}\sum_{i=1}^{n}q_{\boldsymbol{\phi}}(\mathbf{z}|\mathbf{x}_i)$$

- $\mathbb{I}_q[\mathbf{x},\mathbf{z}]$ is the mutual information between $\mathbf{x}$ and $\mathbf{z}$ under the data distribution $p_{\mathrm{data}}(\mathbf{x})$ and $q_{\boldsymbol{\phi}}(\mathbf{z}|\mathbf{x})$.
- The first term encourages $q_{\mathrm{agg},\boldsymbol{\phi}}(\mathbf{z})$ to match the prior $p(\mathbf{z})$.
- The second term reduces the information about $\mathbf{x}$ encoded in $\mathbf{z}$.

Hoffman M. D., Johnson M. J. ELBO Surgery: Yet Another Way to Carve Up the Variational Evidence Lower Bound, 2016

ELBO Surgery

$$\frac{1}{n}\sum_{i=1}^{n} D_{\mathrm{KL}}(q_{\boldsymbol{\phi}}(\mathbf{z}|\mathbf{x}_i)\|p(\mathbf{z})) = D_{\mathrm{KL}}(q_{\mathrm{agg},\boldsymbol{\phi}}(\mathbf{z})\|p(\mathbf{z})) + \mathbb{I}_q[\mathbf{x},\mathbf{z}].$$

Proof

$$\frac{1}{n}\sum_{i=1}^{n} D_{\mathrm{KL}}(q_{\boldsymbol{\phi}}(\mathbf{z}|\mathbf{x}_i)\|p(\mathbf{z})) = \frac{1}{n}\sum_{i=1}^{n}\int q_{\boldsymbol{\phi}}(\mathbf{z}|\mathbf{x}_i)\log\frac{q_{\boldsymbol{\phi}}(\mathbf{z}|\mathbf{x}_i)}{p(\mathbf{z})}d\mathbf{z}$$

$$= \frac{1}{n}\sum_{i=1}^{n}\int q_{\boldsymbol{\phi}}(\mathbf{z}|\mathbf{x}_i)\log\frac{q_{\mathrm{agg},\boldsymbol{\phi}}(\mathbf{z})q_{\boldsymbol{\phi}}(\mathbf{z}|\mathbf{x}_i)}{p(\mathbf{z})q_{\mathrm{agg},\boldsymbol{\phi}}(\mathbf{z})}d\mathbf{z}$$

$$= \int \frac{1}{n}\sum_{i=1}^{n} q_{\boldsymbol{\phi}}(\mathbf{z}|\mathbf{x}_i)\log\frac{q_{\mathrm{agg},\boldsymbol{\phi}}(\mathbf{z})}{p(\mathbf{z})}d\mathbf{z} + \frac{1}{n}\sum_{i=1}^{n}\int q_{\boldsymbol{\phi}}(\mathbf{z}|\mathbf{x}_i)\log\frac{q_{\boldsymbol{\phi}}(\mathbf{z}|\mathbf{x}_i)}{q_{\mathrm{agg},\boldsymbol{\phi}}(\mathbf{z})}d\mathbf{z}$$

$$= D_{\mathrm{KL}}(q_{\mathrm{agg},\boldsymbol{\phi}}(\mathbf{z})\|p(\mathbf{z})) + \frac{1}{n}\sum_{i=1}^{n} D_{\mathrm{KL}}(q_{\boldsymbol{\phi}}(\mathbf{z}|\mathbf{x}_i)\|q_{\mathrm{agg},\boldsymbol{\phi}}(\mathbf{z}))$$

$$\mathbb{I}_q[\mathbf{x},\mathbf{z}] = \frac{1}{n}\sum_{i=1}^{n} D_{\mathrm{KL}}(q_{\boldsymbol{\phi}}(\mathbf{z}|\mathbf{x}_i)\|q_{\mathrm{agg},\boldsymbol{\phi}}(\mathbf{z})).$$

ELBO Surgery

Revisiting the ELBO

$$\frac{1}{n}\sum_{i=1}^{n}\mathcal{L}_{\phi,\theta}(\mathbf{x}_i) = \frac{1}{n}\sum_{i=1}^{n}\left[\mathbb{E}_{q_\phi(\mathbf{z}|\mathbf{x}_i)}\log p_\theta(\mathbf{x}_i|\mathbf{z}) - D_{\mathrm{KL}}(q_\phi(\mathbf{z}|\mathbf{x}_i)\|p(\mathbf{z}))\right]$$

$$= \underbrace{\frac{1}{n}\sum_{i=1}^{n}\mathbb{E}_{q_\phi(\mathbf{z}|\mathbf{x}_i)}\log p_\theta(\mathbf{x}_i|\mathbf{z})}_{\text{Reconstruction Loss}} - \underbrace{\mathbb{I}_q[\mathbf{x},\mathbf{z}]}_{\text{Mutual Information}} - \underbrace{D_{\mathrm{KL}}(q_{\mathrm{agg},\phi}(\mathbf{z})\|p(\mathbf{z}))}_{\text{Marginal KL}}$$

> The **prior distribution** $p(\mathbf{z})$ only appears in the **Marginal KL** term.

Optimal VAE Prior

$$D_{\mathrm{KL}}(q_{\mathrm{agg},\phi}(\mathbf{z})\|p(\mathbf{z})) = 0 \quad \Leftrightarrow \quad p(\mathbf{z}) = q_{\mathrm{agg},\phi}(\mathbf{z}) = \frac{1}{n}\sum_{i=1}^{n} q_\phi(\mathbf{z}|\mathbf{x}_i).$$

> Hence, the **optimal prior** $p(\mathbf{z})$ is the **aggregated variational posterior** $q_{\mathrm{agg},\phi}(\mathbf{z})$.

Hoffman M. D., Johnson M. J. ELBO Surgery: Yet Another Way to Carve Up the Variational Evidence Lower Bound, 2016

Marginal KL

$$D_{\mathrm{KL}}(q_{\mathrm{agg},\boldsymbol{\phi}}(\mathbf{z})\|p(\mathbf{z}))$$

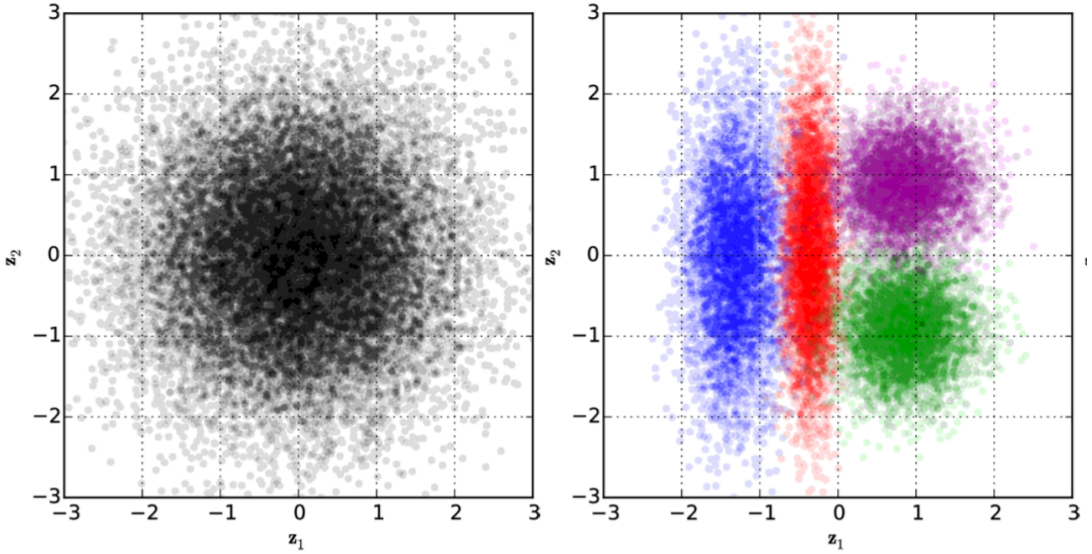
Training

The decoder receives $\mathbf{z} \sim q_{\boldsymbol{\phi}}(\mathbf{z}|\mathbf{x}_i)$; over the training set, $\mathbf{z} \sim q_{\mathrm{agg},\boldsymbol{\phi}}(\mathbf{z})$.

Sampling (generation)

The decoder receives $\mathbf{z} \sim p(\mathbf{z})$, then generates $\mathbf{x} \sim p_{\boldsymbol{\theta}}(\mathbf{x}|\mathbf{z})$.

- $q_{\boldsymbol{\phi}}(\mathbf{z}|\mathbf{x}) = \mathcal{N}(\boldsymbol{\mu}_{\boldsymbol{\phi}}(\mathbf{x}), \boldsymbol{\sigma}^2_{\boldsymbol{\phi}}(\mathbf{x}))$ is unimodal.
- A mismatch between the prior and the aggregated posterior can degrade sample quality. Matching these distributions does not by itself eliminate blurriness.

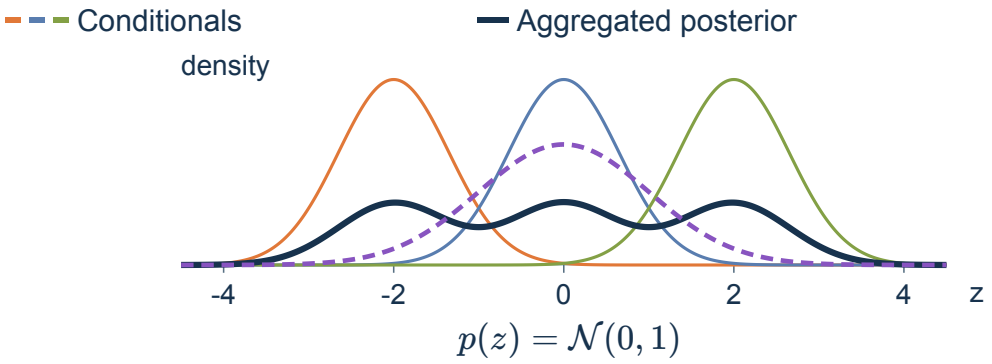


(a) Prior distribution (b) Posteriors in standard VAE

Matching the Prior: Which KL Vanishes?

$$\frac{1}{n}\sum_{i=1}^{n} D_{\mathrm{KL}}(q_\phi(\mathbf{z}|\mathbf{x}_i)\|p(\mathbf{z})) = D_{\mathrm{KL}}(q_{\mathrm{agg},\phi}(\mathbf{z})\|p(\mathbf{z})) + \mathbb{I}_q[\mathbf{x},\mathbf{z}].$$

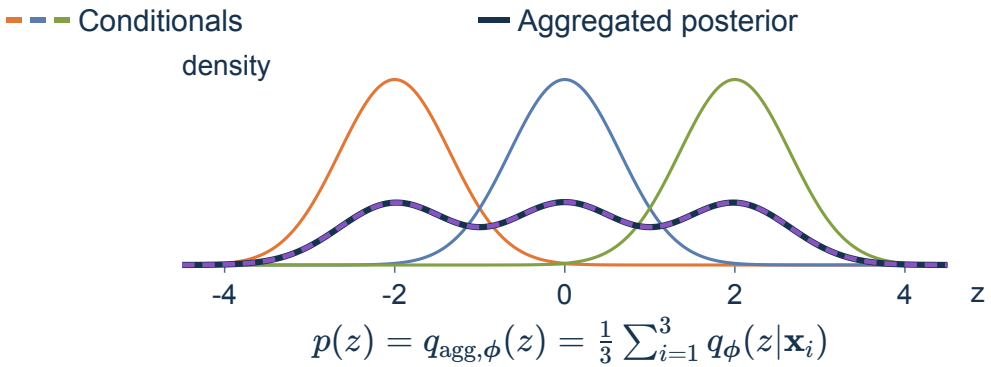
$$q_\phi(z|\mathbf{x}_i) = \mathcal{N}(\mu_i, 0.65^2), \quad (\mu_1,\mu_2,\mu_3) = (-2,0,2), \quad p_{\mathrm{data}}(\mathbf{x}_i) = 1/3$$



$p(z) = \mathcal{N}(0,1)$

Marginal KL	**0.584**
Mutual information	**0.892**
Mean conditional KL	**1.475**

All values in nats; all bars use the same scale.



$p(z) = q_{\mathrm{agg},\phi}(z) = \frac{1}{3}\sum_{i=1}^{3} q_\phi(z|\mathbf{x}_i)$

Marginal KL	**0.000**
Mutual information	**0.892**
Mean conditional KL	**0.892**

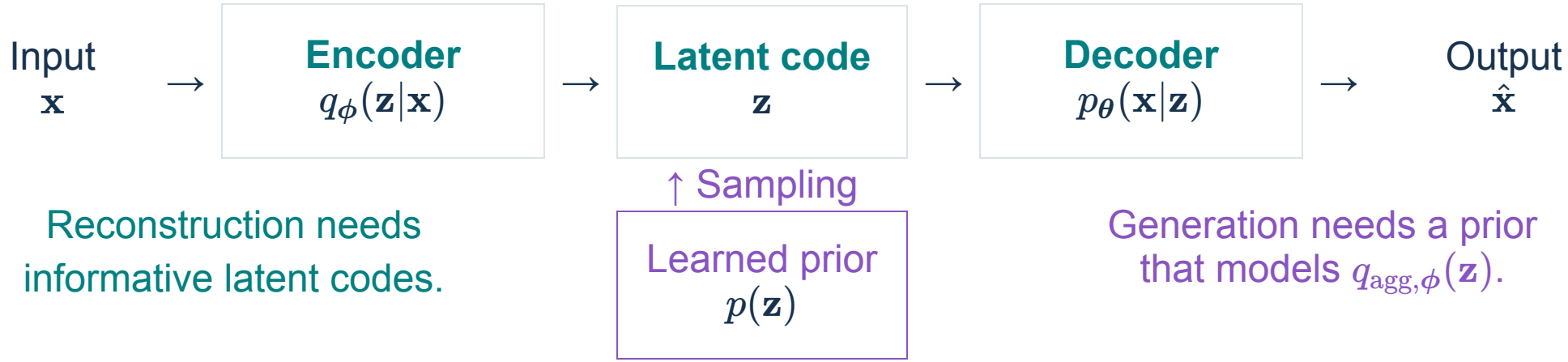
All values in nats; all bars use the same scale.

Matching the prior removes **Marginal KL**. The **mutual information** and encoder remain unchanged.

ELBO Surgery in Modern Latent Models

Jointly learn an **autoencoder** and a **generative prior**.

$$\text{Average conditional KL} = \text{Marginal KL} + \text{Mutual Information}.$$



> Latents that **reconstruct well** may be **hard for the prior to model**.

GenFirst (2026) studies this balance when the autoencoder and generative prior are trained jointly.

Hoffman M. D., Johnson M. J. ELBO Surgery, 2016.
Zheng G. et al. GenFirst: Generation Before Reconstruction for Stable End-to-End Latent Generative Modeling, 2026.

From ELBO to Squared Error

Fix the encoder $q_{\boldsymbol{\phi}}(\mathbf{z}|\mathbf{x})$, prior $p(\mathbf{z})$ and decoder variance $\sigma^2 > 0$; optimize only $\boldsymbol{\theta}$.

$$\mathbb{E}_{p_{\text{data}}(\mathbf{x})}\mathcal{L}_{\boldsymbol{\phi},\boldsymbol{\theta}}(\mathbf{x}) = \mathbb{E}_{p_{\text{data}}(\mathbf{x})q_{\boldsymbol{\phi}}(\mathbf{z}|\mathbf{x})}\log p_{\boldsymbol{\theta}}(\mathbf{x}|\mathbf{z}) - \underbrace{\mathbb{E}_{p_{\text{data}}(\mathbf{x})}D_{\text{KL}}(q_{\boldsymbol{\phi}}(\mathbf{z}|\mathbf{x})\|p(\mathbf{z}))}_{\text{constant w.r.t. } \boldsymbol{\theta}}.$$

The **conditional KL is constant in $\boldsymbol{\theta}$**, not necessarily zero. At $p = q_{\text{agg},\boldsymbol{\phi}}$:

$$\underbrace{D_{\text{KL}}(q_{\text{agg},\boldsymbol{\phi}}(\mathbf{z})\|p(\mathbf{z}))}_{\text{Marginal KL}} = 0, \qquad \underbrace{\frac{1}{n}\sum_{i=1}^{n} D_{\text{KL}}(q_{\boldsymbol{\phi}}(\mathbf{z}|\mathbf{x}_i)\|p(\mathbf{z}))}_{\text{Average conditional KL}} = \mathbb{I}_q[\mathbf{x},\mathbf{z}].$$

Gaussian Decoder: Squared Reconstruction Error

$$p_{\boldsymbol{\theta}}(\mathbf{x}|\mathbf{z}) = \mathcal{N}(\boldsymbol{\mu}_{\boldsymbol{\theta}}(\mathbf{z}), \sigma^2\mathbf{I}) \quad \Rightarrow \quad \log p_{\boldsymbol{\theta}}(\mathbf{x}|\mathbf{z}) = -\frac{\|\mathbf{x} - \boldsymbol{\mu}_{\boldsymbol{\theta}}(\mathbf{z})\|_2^2}{2\sigma^2} + \text{const}.$$

$$\arg\max_{\boldsymbol{\theta}} \mathbb{E}_{p_{\text{data}}(\mathbf{x})}\mathcal{L}_{\boldsymbol{\phi},\boldsymbol{\theta}}(\mathbf{x}) = \arg\min_{\boldsymbol{\theta}} \mathbb{E}_{p_{\text{data}}(\mathbf{x})q_{\boldsymbol{\phi}}(\mathbf{z}|\mathbf{x})}\|\mathbf{x} - \boldsymbol{\mu}_{\boldsymbol{\theta}}(\mathbf{z})\|_2^2.$$

Why are VAE Generations Blurry?

Keep the encoder, prior and Gaussian decoder variance fixed. Maximizing ELBO gives

$$\arg\min_{\boldsymbol{\theta}} \mathbb{E}_{p_{\text{data}}(\mathbf{x})q_{\boldsymbol{\phi}}(\mathbf{z}|\mathbf{x})} \|\mathbf{x} - \boldsymbol{\mu}_{\boldsymbol{\theta}}(\mathbf{z})\|_2^2.$$

Optimal Decoder Mean

The optimal decoder mean is the **conditional expectation**:

$$\boldsymbol{\mu}^*(\mathbf{z}) = \mathbb{E}_{q_{\boldsymbol{\phi}}(\mathbf{x}|\mathbf{z})}[\mathbf{x}] = \frac{\mathbb{E}_{p_{\text{data}}(\mathbf{x})}[q_{\boldsymbol{\phi}}(\mathbf{z}|\mathbf{x}) \cdot \mathbf{x}]}{\mathbb{E}_{p_{\text{data}}(\mathbf{x})}[q_{\boldsymbol{\phi}}(\mathbf{z}|\mathbf{x})]}$$

Blurriness from Averaging

- $p_{\text{data}}(\mathbf{x})$ and $q_{\boldsymbol{\phi}}(\mathbf{z}|\mathbf{x})$ give us the aggregated posterior $q_{\text{agg},\boldsymbol{\phi}}(\mathbf{z})$.
- If $\mathbf{x} \neq \mathbf{x}'$ map to overlapping latent regions, $\boldsymbol{\mu}^*(\mathbf{z})$ averages over unrelated inputs.
- This leads to **blurry, non-distinct outputs**.

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Discrete VAE Latent Representations

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03 Likelihood-Free Learning

Discrete VAE Latents

Motivation

- VAE has used **continuous** latent variables $\mathbf{z}$.
- For some modalities, **discrete** representations $\mathbf{z}$ may be a more natural choice.
- Advanced autoregressive models are highly effective for distributions over discrete variables (for example, transformers process discrete tokens).

ELBO

$$\mathcal{L}_{\boldsymbol{\phi},\boldsymbol{\theta}}(\mathbf{x}) = \mathbb{E}_{q_{\boldsymbol{\phi}}(\mathbf{z}|\mathbf{x})} \log p_{\boldsymbol{\theta}}(\mathbf{x}|\mathbf{z}) - D_{\mathrm{KL}}(q_{\boldsymbol{\phi}}(\mathbf{z}|\mathbf{x})||p(\mathbf{z})) \to \max_{\boldsymbol{\phi},\boldsymbol{\theta}}.$$

- Apply the reparametrization trick to obtain unbiased gradients.
- Use Gaussian distributions for $q_{\boldsymbol{\phi}}(\mathbf{z}|\mathbf{x})$ and $p(\mathbf{z})$ to compute the KL analytically.

Discrete VAE Latents

Assumptions

- Let $c \sim \mathrm{Cat}(\boldsymbol{\pi})$, where

$$\boldsymbol{\pi} = (\pi_1, \ldots, \pi_K), \quad \pi_k = P(c = k), \quad \sum_{k=1}^{K} \pi_k = 1.$$

- For tokenizer training, fix $p(c) = \mathrm{Uniform}\{1, \ldots, K\}$.

ELBO

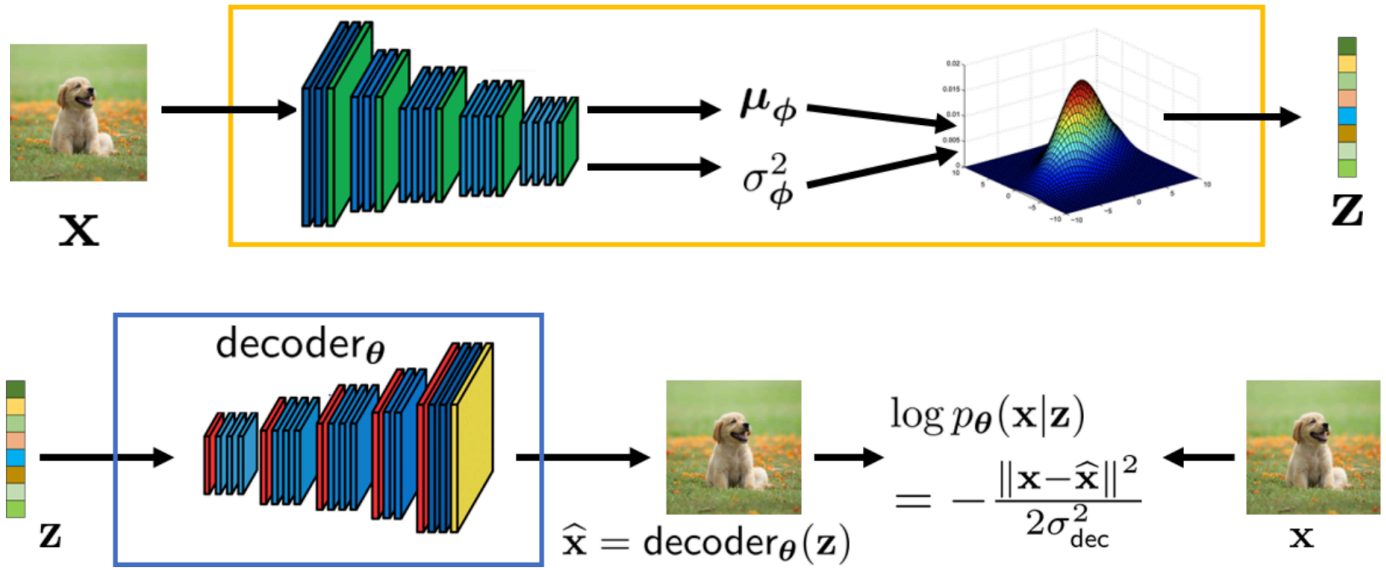
$$\mathcal{L}_{\boldsymbol{\phi},\boldsymbol{\theta}}(\mathbf{x}) = \mathbb{E}_{q_{\boldsymbol{\phi}}(c|\mathbf{x})} \log p_{\boldsymbol{\theta}}(\mathbf{x}|c) - D_{\mathrm{KL}}(q_{\boldsymbol{\phi}}(c|\mathbf{x}) \| p(c)) \to \max_{\boldsymbol{\phi},\boldsymbol{\theta}}.$$

$$\begin{aligned} D_{\mathrm{KL}}(q_{\boldsymbol{\phi}}(c|\mathbf{x}) \| p(c)) &= \sum_{k=1}^{K} q_{\boldsymbol{\phi}}(k|\mathbf{x}) \log \frac{q_{\boldsymbol{\phi}}(k|\mathbf{x})}{p(k)} \\ &= \sum_{k=1}^{K} q_{\boldsymbol{\phi}}(k|\mathbf{x}) \log q_{\boldsymbol{\phi}}(k|\mathbf{x}) - \sum_{k=1}^{K} q_{\boldsymbol{\phi}}(k|\mathbf{x}) \log p(k) \\ &= -\mathrm{H}(q_{\boldsymbol{\phi}}(c|\mathbf{x})) + \log K. \end{aligned}$$

Discrete VAE Latents

$$\mathcal{L}_{\phi,\theta}(\mathbf{x}) = \mathbb{E}_{q_\phi(c|\mathbf{x})} \log p_\theta(\mathbf{x}|c) + \mathrm{H}(q_\phi(c|\mathbf{x})) - \log K \to \max_{\phi,\theta}.$$

- The encoder should output a discrete distribution $q_\phi(c|\mathbf{x})$.
- We need an analogue of the reparametrization trick for discrete $q_\phi(c|\mathbf{x})$.
- The decoder $p_\theta(\mathbf{x}|c)$ must take a discrete random variable c as input.



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Vector Quantization

A **codebook** contains K vectors: $\{\mathbf{e}_k\}_{k=1}^{K}$, with $\mathbf{e}_k \in \mathbb{R}^L$.

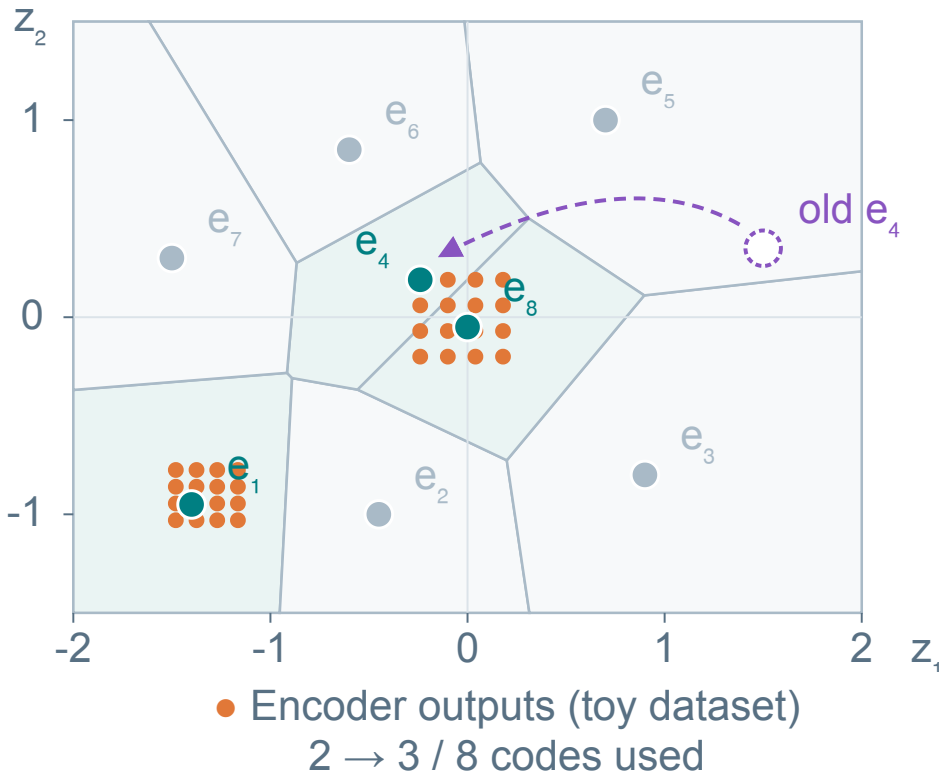
$$\mathbf{z}_q = \mathbf{q}(\mathbf{z}) = \mathbf{e}_{k^*}, \quad \text{where } k^* = \arg\min_k \|\mathbf{z} - \mathbf{e}_k\|.$$

Codebook Collapse

Only a small subset of entries is used across the data.

Restart an Unused Code

Move it to an encoder output; assignments and Voronoi regions change.



Vector Quantized VAE (VQ-VAE)

- The encoder outputs a continuous vector $\mathbf{z}_e = \mathrm{NN}_{e,\boldsymbol{\phi}}(\mathbf{x}) \in \mathbb{R}^L$.
- Quantization **deterministically** maps $\mathbf{z}_e$ to its quantized codebook vector $\mathbf{z}_q$.
- The decoder is conditioned on codebook entries $\mathbf{e}_c$, i.e., via $p_{\boldsymbol{\theta}}(\mathbf{x}|\mathbf{e}_c)$ (instead of $p_{\boldsymbol{\theta}}(\mathbf{x}|c)$).

Deterministic Variational Posterior

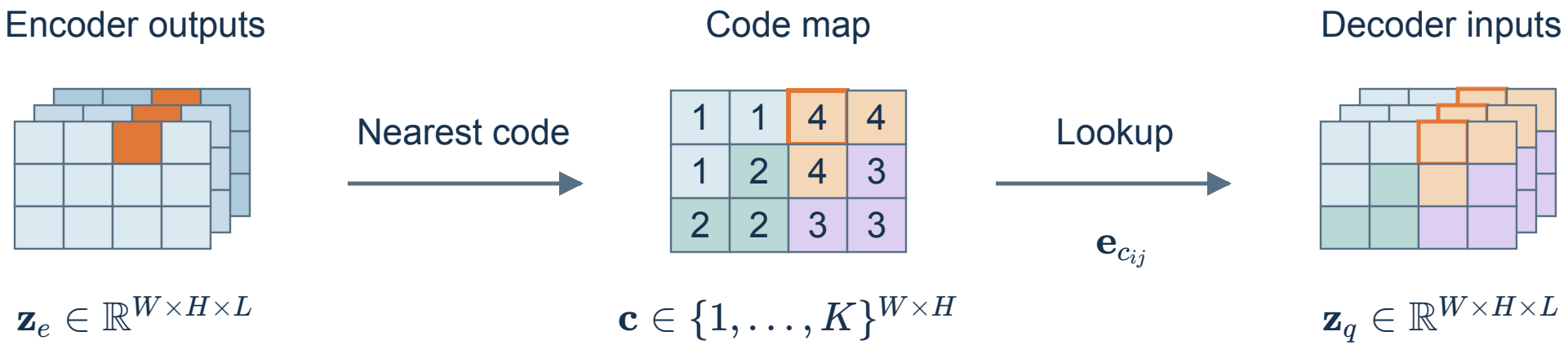
$$q_{\boldsymbol{\phi}}(c = k^*|\mathbf{x}) = \begin{cases} 1, & \text{for } k^* = \arg\min_k \|\mathbf{z}_e - \mathbf{e}_k\|; \\ 0, & \text{otherwise.} \end{cases}$$

$$D_{\mathrm{KL}}(q_{\boldsymbol{\phi}}(c|\mathbf{x})\|p(c)) = -\underbrace{\mathrm{H}(q_{\boldsymbol{\phi}}(c|\mathbf{x}))}_{=0} + \log K = \log K.$$

Constant KL: The aggregated posterior is not pushed toward the uniform prior. We therefore fit a prior to it in a second stage.

Vector Quantized VAE (VQ-VAE): Forward

Quantize each of the $M = WH$ vectors using the **same codebook**.



ELBO

$$\mathcal{L}_{\phi,\theta}(\mathbf{x}) = \mathbb{E}_{q_\phi(\mathbf{c}|\mathbf{x})} \log p_\theta(\mathbf{x}|\mathbf{z}_q(\mathbf{c})) - M \log K = \log p_\theta(\mathbf{x}|\mathbf{z}_q) - M \log K.$$

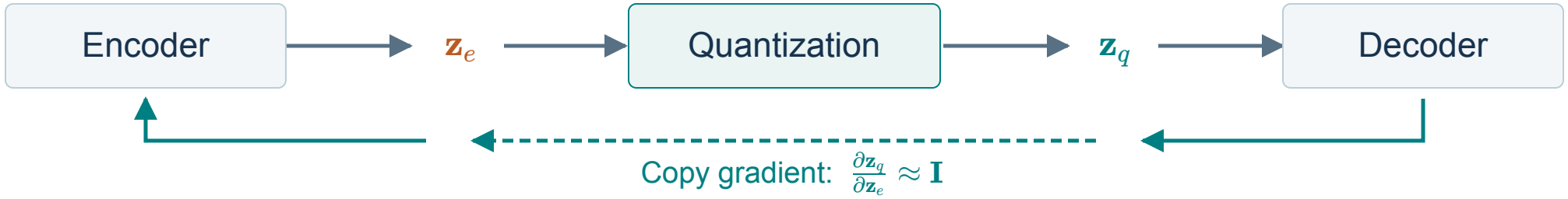
The deterministic map and independent uniform training prior give $D_{\mathrm{KL}} = M \log K$.

Challenge: The $\arg\min$ operation is non-differentiable.

Vector Quantized VAE (VQ-VAE): Backward

ELBO

$$\mathcal{L}_{\phi,\theta}(\mathbf{x}) = \log p_{\theta}(\mathbf{x}|\mathbf{z}_q) - M \log K, \quad \mathbf{z}_{q,m} = \mathbf{e}_{c_m}, \quad m = 1, \dots, M.$$



Straight-Through Gradient Estimator

$$\begin{aligned}
\frac{\partial \log p(\mathbf{x}|\mathbf{z}_q, \boldsymbol{\theta})}{\partial \boldsymbol{\phi}} &= \frac{\partial \log p_{\boldsymbol{\theta}}(\mathbf{x}|\mathbf{z}_q)}{\partial \mathbf{z}_q} \cdot \frac{\partial \mathbf{z}_q}{\partial \boldsymbol{\phi}} \\
&= \frac{\partial \log p_{\boldsymbol{\theta}}(\mathbf{x}|\mathbf{z}_q)}{\partial \mathbf{z}_q} \cdot \frac{\partial \mathbf{z}_q}{\partial \mathbf{z}_e} \cdot \frac{\partial \mathbf{z}_e}{\partial \boldsymbol{\phi}} \\
&\approx \frac{\partial \log p_{\boldsymbol{\theta}}(\mathbf{x}|\mathbf{z}_q)}{\partial \mathbf{z}_q} \cdot \frac{\partial \mathbf{z}_e}{\partial \boldsymbol{\phi}}
\end{aligned}$$

VQ-VAE: Learning a Prior over Codes

The **tokenizer** (encoder + quantizer) produces $M = WH$ codes: $\mathbf{c} = (c_1, \ldots, c_M)$.

Target: Aggregated Posterior (ELBO Surgery)

$$p_{\boldsymbol{\psi}}(\mathbf{c}) \approx q_{\text{agg},\boldsymbol{\phi}}(\mathbf{c}) = \frac{1}{n}\sum_{i=1}^{n} q_{\boldsymbol{\phi}}(\mathbf{c}|\mathbf{x}_i).$$

Training

1. Fix the trained tokenizer and decoder.
2. Encode the training images $\mathbf{x}_i$ into code maps $\mathbf{c}_i$.
3. Fit the **autoregressive prior** (Lecture 1) by maximum likelihood:

$$\sum_{i=1}^{n} \log p_{\boldsymbol{\psi}}(\mathbf{c}_i) = \sum_{i=1}^{n}\sum_{m=1}^{M} \log p_{\boldsymbol{\psi}}(c_{i,m}|c_{i,1}, \ldots, c_{i,m-1}) \to \max_{\boldsymbol{\psi}}.$$

> $p_{\boldsymbol{\psi}}(\mathbf{c}) \to$ **code map** $\to$ **codebook lookup** $\to$ **decoder** $\to \mathbf{x}$.

Vector Quantized VAE-2 (VQ-VAE-2)

VQ-VAE-2 combines code maps at **multiple resolutions**.

Sample Diversity: Learned Autoregressive Priors



VQ-VAE (Proposed) BigGAN deep

Vector Quantized VAE (VQ-VAE): Final algorithm

Training

Stage 1: tokenizer.

1. Sample $\mathbf{x} \sim p_{\text{data}}(\mathbf{x})$.
2. Compute encoder outputs $\mathbf{z}_{e,m} = \text{NN}_{e,\boldsymbol{\phi}}(\mathbf{x})_m$.
3. Compute the quantized representation at each location:
$$c_m = \arg\min_k \|\mathbf{z}_{e,m} - \mathbf{e}_k\|_2, \quad \mathbf{z}_{q,m} = \mathbf{e}_{c_m}.$$
4. Compute the ELBO with the uniform training prior:
$$\mathcal{L}_{\boldsymbol{\phi},\boldsymbol{\theta}}(\mathbf{x}) = \log p_{\boldsymbol{\theta}}(\mathbf{x}|\mathbf{z}_q) - M \log K.$$
5. Compute the loss with codebook and commitment terms:
$$\mathcal{L} = -\mathcal{L}_{\boldsymbol{\phi},\boldsymbol{\theta}}(\mathbf{x}) + \sum_{m=1}^{M} \left\|\text{sg}[\mathbf{z}_{e,m}] - \mathbf{e}_{c_m}\right\|_2^2 + \beta \sum_{m=1}^{M} \left\|\mathbf{z}_{e,m} - \text{sg}[\mathbf{e}_{c_m}]\right\|_2^2.$$
6. Update $\boldsymbol{\phi}$, $\boldsymbol{\theta}$ and $\{\mathbf{e}_k\}$ (straight-through for $\boldsymbol{\phi}$).

Sampling

After **Stage 2: prior training**, fix all parameters.

1. Sample the code map $\mathbf{c} \sim p_{\boldsymbol{\psi}}(\mathbf{c})$ autoregressively.
2. Compute $\mathbf{z}_{q,m} = \mathbf{e}_{c_m}$ at each location.
3. Sample $\mathbf{x} \sim p_{\boldsymbol{\theta}}(\mathbf{x}|\mathbf{z}_q)$.

Outline

01 ELBO Surgery and Optimal VAE Prior

02 Discrete Latent VAEs

Discrete VAE Latent Representations
Vector Quantized VAE (VQ-VAE)

03 Likelihood-Free Learning

Likelihood-Based Models

Poor Likelihood
High-Quality Samples

$$p_1(\mathbf{x}) = \frac{1}{n}\sum_{i=1}^{n}\mathcal{N}(\mathbf{x}|\mathbf{x}_i, \epsilon\mathbf{I})$$

If ϵ is very small, this model produces excellent, sharp samples but achieves poor likelihoods on test data.

High Likelihood
Poor Samples

$$\begin{aligned} p_2(\mathbf{x}) &= 0.01p(\mathbf{x}) + 0.99p_{\text{noise}}(\mathbf{x}) \\ \log\left[0.01p(\mathbf{x}) + 0.99p_{\text{noise}}(\mathbf{x})\right] &\geq \log[0.01p(\mathbf{x})] \\ &= \log p(\mathbf{x}) - \log 100 \end{aligned}$$

This model contains mostly noisy, irrelevant samples; for high dimensions, $\log p(\mathbf{x})$ scales linearly with m.

- Likelihood isn't always a suitable metric for evaluating generative models.
- Sometimes, the likelihood function can't even be computed exactly.

Theis L., Oord A., Bethge M. A Note on the Evaluation of Generative Models, 2015

Likelihood-Free Learning

Motivation

We're interested in approximating the true data distribution $p_{\text{data}}(\mathbf{x})$. Instead of searching over all distributions, let's learn a model $p_{\boldsymbol{\theta}}(\mathbf{x}) \approx p_{\text{data}}(\mathbf{x})$.

Suppose we have two sets of samples:

- $\{\mathbf{x}_i\}_{i=1}^{n_1} \sim p_{\text{data}}(\mathbf{x})$ — real data;
- $\{\mathbf{x}_i\}_{i=1}^{n_2} \sim p_{\boldsymbol{\theta}}(\mathbf{x})$ — generated (fake) data.

Define a discriminative model (classifier):

$$p(y = 1|\mathbf{x}) = P(\mathbf{x} \sim p_{\text{data}}(\mathbf{x})); \quad p(y = 0|\mathbf{x}) = P(\mathbf{x} \sim p_{\boldsymbol{\theta}}(\mathbf{x}))$$

Assumption

The generative model $p_{\boldsymbol{\theta}}(\mathbf{x})$ matches $p_{\text{data}}(\mathbf{x})$ if a discriminative model $p(y|\mathbf{x})$ can't distinguish between them — that is, if $p(y = 1|\mathbf{x}) = 0.5$ for every $\mathbf{x}$.

Generative Adversarial Networks (GAN)

- The more expressive the discriminator $p(y|\mathbf{x})$, the closer we get to the optimal $p_{\boldsymbol{\theta}}(\mathbf{x})$.
- Standard classifiers are trained by minimizing cross-entropy loss $-\mathbb{E}_{\hat{p}(\mathbf{x},y)} \log p(y|\mathbf{x})$ with $\hat{p}(\mathbf{x}, y) = \frac{1}{2}\mathbb{I}_{y=1} p_{\text{data}}(\mathbf{x}) + \frac{1}{2}\mathbb{I}_{y=0} p_{\boldsymbol{\theta}}(\mathbf{x})$.

Cross-Entropy for Discriminator

$$\min_{p(y|\mathbf{x})} \left[-\mathbb{E}_{p_{\text{data}}(\mathbf{x})} \log p(y=1|\mathbf{x}) - \mathbb{E}_{p_{\boldsymbol{\theta}}(\mathbf{x})} \log p(y=0|\mathbf{x})\right]$$

$$\max_{p(y|\mathbf{x})} \left[\mathbb{E}_{p_{\text{data}}(\mathbf{x})} \log p(y=1|\mathbf{x}) + \mathbb{E}_{p_{\boldsymbol{\theta}}(\mathbf{x})} \log p(y=0|\mathbf{x})\right]$$

Generative Model

Suppose $p_{\boldsymbol{\theta}}(\mathbf{x}, \mathbf{z}) = p_{\boldsymbol{\theta}}(\mathbf{x}|\mathbf{z})p(\mathbf{z})$, where $p(\mathbf{z})$ is a base distribution, and $p_{\boldsymbol{\theta}}(\mathbf{x}|\mathbf{z}) = \delta(\mathbf{x} - \mathbf{G}_{\boldsymbol{\theta}}(\mathbf{z}))$ is deterministic.

> **Connection to VAEs:** Both models sample $\mathbf{z} \sim p(\mathbf{z})$. A VAE then samples $\mathbf{x} \sim p_{\boldsymbol{\theta}}(\mathbf{x}|\mathbf{z})$; here, the decoder is **deterministic**: $\mathbf{x} = \mathbf{G}_{\boldsymbol{\theta}}(\mathbf{z})$.

Generative Adversarial Networks (GAN)

Cross-Entropy for Discriminative Model

$$\max_{p(y|\mathbf{x})} \left[\mathbb{E}_{p_{\text{data}}(\mathbf{x})} \log p(y=1|\mathbf{x}) + \mathbb{E}_{p_{\boldsymbol{\theta}}(\mathbf{x})} \log p(y=0|\mathbf{x})\right]$$

- **Discriminator:** A classifier $p_{\boldsymbol{\phi}}(y=1|\mathbf{x}) = D_{\boldsymbol{\phi}}(\mathbf{x}) \in [0,1]$, distinguishing real and generated samples. The discriminator aims to **maximize** cross-entropy.
- **Generator:** The generative model $\mathbf{x} = \mathbf{G}_{\boldsymbol{\theta}}(\mathbf{z}),\ \mathbf{z} \sim p(\mathbf{z})$, seeks to fool the discriminator. The generator aims to **minimize** cross-entropy.

GAN Objective

$$\min_G \max_D \left[\mathbb{E}_{p_{\text{data}}(\mathbf{x})} \log D(\mathbf{x}) + \mathbb{E}_{p_{\boldsymbol{\theta}}(\mathbf{x})} \log(1 - D(\mathbf{x}))\right]$$

$$\min_G \max_D \left[\mathbb{E}_{p_{\text{data}}(\mathbf{x})} \log D(\mathbf{x}) + \mathbb{E}_{p(\mathbf{z})} \log(1 - D(\mathbf{G}(\mathbf{z})))\right]$$

The last step is LOTUS (Law of the Unconscious Statistician): use $\mathbf{x} = \mathbf{G}(\mathbf{z})$ to rewrite the expectation over $p_{\boldsymbol{\theta}}(\mathbf{x})$ as an expectation over $p(\mathbf{z})$.

Summary

- ELBO surgery gives insights into the prior’s influence in VAEs; the optimal prior is the aggregated variational posterior.
- With a Gaussian decoder, averaging over distinct inputs that share a latent region can produce blurry outputs even when the prior matches the aggregated posterior.
- VQ-VAE learns discrete codes; a second stage fits their aggregated posterior with a generative prior.
- The straight-through gradient estimator allows gradients to pass as if quantization were an identity operation during backpropagation.
- Likelihood is not always a suitable metric for evaluating generative models; likelihood-free learning motivates the use of discriminative models to match distributions.
- GANs formulate distribution matching as a minimax game between a generator and a discriminator.